

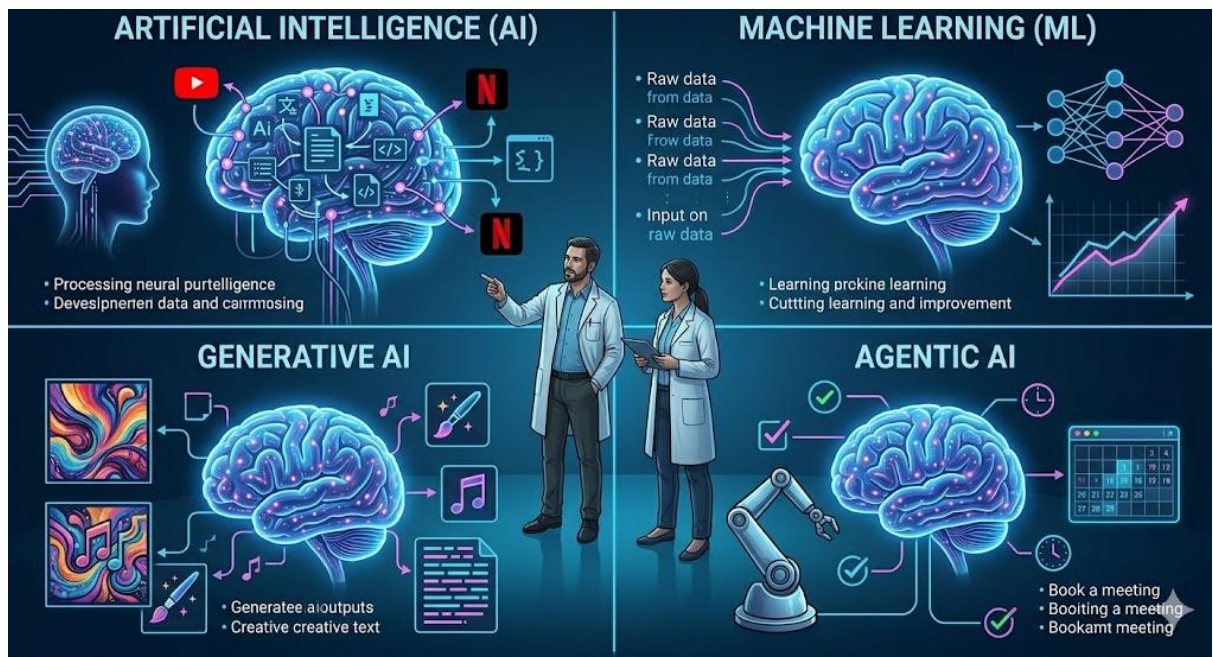
1 . What is Artificial Intelligence

- **Artificial Intelligence (AI)** is a branch of computer science where machines are made to think and act like humans.
- AI means making computers or machines smart enough to learn, understand, and make decisions.
- AI works by using data and rules so that machines can recognize patterns, solve problems, and improve over time.
- **Ex:** Youtube , Netflix etc

2 . Difference between AI, Machine Learning, Generative AI, and Agentic AI.

AI	Machine Learning	Generative AI	Agentic AI
Thinks like a human.	Learns from data.	Creates new content.	Does tasks by itself.
Solves problems.	Finds patterns.	Generates text and images.	Makes decisions.
Makes machines smart.	Improves with experience.	Produces new ideas.	Takes actions automatically.
Example: Siri.	Example: YouTube recommendations.	Example: ChatGPT.	Example: AI assistant booking meetings.

The Evolution of AI: From General Concepts to Agentic Systems



3 . Explain any three open-source AI models available on Hugging Face.

There are three Well-Known Open-Source AI Models on Hugging Face. They are:

- **Llama 3**
- **Mistral 7B**
- **BERT**

Llama 3: Developed by Meta. It is used for chatbots, text generation, coding, and question answering.

Mistral 7B: Developed by Mistral AI. It is known for fast and efficient text generation with strong performance.

BERT: Developed by Google. It is widely used for understanding text, sentiment analysis, and language processing tasks.

4 . What is the difference between traditional programming and AI-assisted development?

Traditional Programming	AI-Assisted Development
Programmer writes all the code manually.	AI helps generate code and suggestions.
Works based on fixed rules.	Uses AI to assist in coding tasks.
Takes more time to develop programs.	Speeds up development process.
Debugging is done manually.	AI can help find and fix errors.
Requires more human effort.	Reduces human effort and increases productivity.
Example: Writing code from scratch.	Example: Using GitHub Copilot

5 . Explain AI hallucinations with one example.

AI hallucinations: These are situations where an AI system generates information that is incorrect, false, or imaginary but presents it as if it is true. This happens when the AI predicts answers based on patterns in data rather than verifying facts.

AI hallucination means the AI creates a wrong answer and presents it confidently as a correct fact.

Example:

If a user asks, *"Who invented the smartphone?"* and the AI answers *"Thomas Edison invented the smartphone,"* the response is incorrect and is called an AI hallucination.